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An extended ISM and MICMAC method under neutrosophic environment

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Abstract

Purpose – ISM technique is one of the tools of interest in soft operations research. The soft nature of this technique has made inevitable use of indeterminacy theories. The present research attempts to develop ISM technique and MICMAC analysis in a neutrosophic space due to the complexity and uncertainty of the decision-making environment.

Design/methodology/approach – In this study, single-valued triangular neutrosophic numbers is used to develop Neutrosophic ISM (NISM) and Neutrosophic MICMAC (NMICMAC). First, the general algorithm of NISM and NMICMAC is provided. In the following, the complete description of NISM steps including level value determination, Factor Leveling Algorithm and NISM digraph algorithm are presented. Finally, a description of the NMICMAC steps is described.

Findings – An illustrative example – supplier selection problem – is given to verify the effectiveness of the proposed method and in the discussion section; the comparison and analysis of different aspects of the NISM with the previous methods are discussed.

Originality/value – In this study, NISM and NMICMAC are presented for the first time, so that each pairwise comparison judgment is provided as single valued triangular neutrosophic numbers. The development of the model is continued until the final stages of calculations with neutrosophic numbers, and only in the final stage the results are crispy presented. In addition, not only the factors of process are leveled, but at each level the factors are lined up and their importance is determined.

Keywords Neutrosophic set, Single value triangular neutrosophic number (SVTrNN), ISM, MICMAC

Paper type Research paper

1. Introduction

Interpretive structural modeling (ISM) was first introduced by Warfield (1974a). ISM is one of the multi-criteria decision-making methods (MCDM) (Jharkharia and Shankar, 2005; Khatwani and Prakash, 2015) which identifies complex relationships between different elements of a system and then organizes them into a hierarchical and structured model of factors considering expert opinions and pair comparisons (Jain and Raj, 2021). ISM is a combination of digraphs theory, matrix operation theory and expert elicitation method aided calculation for better understanding and interpreting the interrelationships between various elements (Phogat and Gupta, 2018). This method is interpretive in that the interrelations between factors are identified and interpreted based on the judgments of an expert group. The modeling is structural in such a way that the interrelations between factors are structured and presented by a relationship digraph (Singh and Kant, 2008). In this method, the relationship matrix is obtained according to the expert opinion.

Additional to the interpretive structural model, MICMAC matrix is usually developed to show the degree of interdependence and mutual influence between factors. MICMAC is



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determinant in the process of decision-making by identifying dominant and fundamental factors. MICMAC was developed to analyze the invisible relationship amongst factors of the ISM model (Singh and Agrawal, 2021). In terms of driving and dependence power, system items are classified into four categories including autonomous, dependent, linkage and driver factors (Aloini *et al.*, 2012; Baykasoglu and Gölçük, 2017).

In conventional ISM and MICMAC approaches, the interrelations between system elements were determined on the assumption that only one binary relation exists between every two factors (Wang *et al.*, 2018). However, decision-makers usually prefer to use linguistic expressions to determine the interrelation between factors due to lack of information or partial knowledge (Biswas *et al.*, 2016). In recent studies, special attention has been paid to verbal uncertainty under the decision-making condition since fuzzy theory proposed by Zadeh (1965). Instead of showing a binary relation in terms of 0 and 1s, fuzzy numbers with a membership degree in the fuzzy set $[0, 1]$ are used to show the degree of relationship between factors. Today, many researchers integrate ISM and fuzzy MICMAC to be applied in various fields such as supply chain management, logistics risk management and knowledge management (Ali *et al.*, 2013; Bhosale and Kant, 2016; Jain and Raj, 2021; Khan *et al.*, 2021; Shanker and Barve, 2021).

However, only the degree of membership (truth degree), neither the degree of non-membership (falsity) nor the degree of indeterminacy, is of importance in fuzzy sets (Abdel-Basset *et al.*, 2018). Therefore, intuitionistic and interval intuitionistic fuzzy sets have been developed to overcome this deficiency. Atanassov (1986) introduced intuitionistic fuzzy sets (IFSs) with membership and non-membership degrees. In this kind of set, as the total degree of membership and non-membership is less than 1, these sets are not able to cope with indeterminacy and contradiction of real conditions.

Based on neutrosophic logic, Smarandache (1999) introduced the concept of neutrosophic sets to deal with indeterminant and contradictory information. These sets are able to show ambiguity, uncertainty, imprecision, vagueness, incompleteness, redundancy, inconsistency and contradiction in human decision-making concerning the incomplete information received from the real world (Ye, 2014a). In a neutrosophic set, not only truth membership function (T) but also indeterminacy membership function (I) and falsity membership function (F) are applied. These sets are a generalization of definite, fuzzy, interval fuzzy, intuitionistic fuzzy and interval intuitionistic fuzzy sets (Biswas *et al.*, 2016). As the direct use of neutrosophic sets was practically difficult, single-valued neutrosophic sets (SVNSs) have been developed in science and engineering so as to be used in practice Wang *et al.* (2010) proposed these sets with respect to the degree of truth, indeterminacy and falsity.

Due to the importance of uncertainty in the process of multi-criteria decision making, many researchers have used neutrosophic sets to expand MCDM methods. Biswas *et al.* (2016) employed TOPSIS method in a single-valued neutrosophic environment. Stanujkic *et al.* (2017) proposed MULTIMOORA approach employing single-valued neutrosophic sets. After 2015, there was a growing use of neutrosophic sets in various disciplines such as MCDM, resulting in a variety of new methods including TOPSIS, WASPA, COPRAS, SWARA, VIKOR, ELECTRE, TODIM, QUALIFLEX, GRA, MULTIMOORA and PROMETHE in all of which neutrosophic sets have been used (Hezam *et al.*, 2022; Peng and Dai, 2020; Wang *et al.*, 2019; Zhang *et al.*, 2023; Zhao *et al.*, 2023). These studies have shown that using neutrosophic sets is a suitable means for expressing opinions, yielding closer results to reality.

Although there have been some studies related to NISM and NMICMAC, the literature on this topic is limited to a small number of researches. Karadayi-Usta (2022) and Dohale *et al.* (2023) are the only relevant studies that were found by searching the databases. These studies examined ISM and MICMAC in the neutrosophic space. But in these two studies only experts' judgments are considered as neutrosophic sets and these conditions are not involved in the process of algorithms and this has led to the loss of neutrosophic processing and considering ambiguity.

This work fills the gap in the literature by providing a new approach to decision-making under uncertainty our work proposes an extension of the methodology under neutrosophic environment as NISM and NMICMAC. The aim of the current research is to achieve a more accurate and realistic hierarchy and, as a result, a better interpretation of the relationships between system factors. In this research, we try to develop ISM, using neutrosophic input, process and output, and the leveling of factors has been presented for the first time. Given that triangular numbers and linguistic term are closely associated, single value triangular neutrosophic numbers have been used to determine the interrelationships.

According to the gap in the literature, the main research question is as follows:

How can ISM and MICMAC techniques be extended based on neutrosophic assumptions? To achieve answers to the questions, this research highlights the following:

- (1) The ISM and MICMAC method in a neutrosophic space is developed for the purpose of applying it in real-life problems.
- (2) Single valued triangular neutrosophic numbers considering the degree of truth, indeterminacy and falsity related to the linguistic variables is used to developed neutrosophic ISM (NISM) and neutrosophic MICMAC (NMICMAC).
- (3) The general algorithm of NISM and NMICMAC as well as the algorithms of factor leveling and digraph were also provided with the aim of maintaining neutrosophic conditions until the end of the decision-making process
- (4) An illustrative example in supply chain management – supplier selection problem-is given to show the usage of the developed methods.
- (5) The proposed model is discussed with the description on its advantageous aspects.

The rest of this article is arranged as follows. [Section 2](#) provides a comprehensive view of the related works on neutrosophic sets. Some definitions and operations on neutrosophic sets are provided in [Section 3](#). In [Sections 4 and 5](#), ISM and MICMAC are introduced. With the help of provided definitions, both presented techniques are then developed into a neutrosophic space in [sections 6 and 7](#). In [Section 8](#), the application of the developed techniques is shown drawing on a numerical example. Finally, some discussion and conclusions are made in [Section 9](#) and [10](#).

2. Literature

In traditional logic, precise symbols are often used, which fail to capture the complexities of reality. The relationship between accuracy and uncertainty has always been a challenge for decision-makers. To address this issue, [Zadeh \(1965\)](#) introduced fuzzy logic, which expanded uncertainty and ambiguity into the formal system of mathematical logic. Later, [Smarandache \(1999\)](#) proposed neutrosophy, a new branch of philosophy that studies the origin, nature and scope of neutrality, as well as its interaction with various intellectual spectrums. Neutrosophy is a multiple-valued logic that includes classical logic, fuzzy logic and imprecise probability. It is closer to human rationality than traditional logic because it accounts for the imprecision of knowledge or linguistic imprecision created by several observers ([Das et al., 2020](#)). In neutrosophic theory, each event has degrees of truth, falsity and indeterminacy, which are considered independently of one another. Neutrosophic sets have garnered attention from researchers due to their ability to describe a wide range of domains in real life while effectively managing ambiguity. A review of the specialized literature reveals that neutrosophic sets are rapidly expanding in both quantitative and qualitative research.

Researchers have proposed several approaches for applying neutrosophic sets in decision-making scenarios. For example, [Rivieccio \(2008\)](#) critically reviewed the relationship between neutrosophic logics and existing frameworks. [Zhang et al. \(2014\)](#) proposed an interval neutrosophic set for use in decision-making scenarios. [Ye \(2014b\)](#) defined Hamming and Euclidean distances between neutrosophic sets and proposed similarity measures based on these distances. [Pamučar and Božanić \(2019\)](#) proposed the multi-attributive border approximation area comparison (MABAC) method, which uses neutrosophic sets for prioritizing decision-makers' risks. Other researchers have used neutrosophic numbers to solve multiple attribute decision-making (MADM) problems. For instance, [Pouresmaeil et al. \(2017\)](#) used neutrosophic numbers to find decision-makers' weights and rank options based on Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) and VIKOR (VlseKriterijumska Optimizacija I Kompromisno Resenje) methods. [Ye \(2015\)](#) introduced trapezoidal neutrosophic sets, developed some operators based on them and applied them to MADM problems. [Tan and Zhang \(2017\)](#) developed the TOPSIS method using trapezoidal fuzzy sets for decision-making under ambiguity. Moreover, researchers have proposed novel concepts in the neutrosophic arena. For instance, [Nancy Garg \(2016\)](#) introduced new scoring functions for ranking neutrosophic numbers. [Deli et al. \(2015\)](#) defined bipolar neutrosophic sets and introduced some related operators. As the literature shows, neutrosophic sets are still evolving, and new ideas are emerging in this area.

Recently, [Peng and Dai \(2020\)](#) in a study investigated a bibliometric analysis of neutrosophic set between 1998 to 2017. [Table 1](#) shows a part of their study in relation to neutrosophic MCDM.

A review of this related literature shows that despite the increasing use of neutrosophic sets in MCDM methods; the application of NISM and NMICMAC has been addressed in less research. A review of the subject literature in various databases such as Scopus, Web of Science and Google Scholar shows the existence of only two works related to NISM.

[Karadayi-Usta \(2022\)](#) proposed a neutrosophic ISM to examine the relationship of the medical tourism services' barriers. In this study, like many researches related to fuzzy ISM, neutrosophic logic sets were used only in the initial stages and to achieve the reachability matrix.

MCDM methods	Frequency of application
Aggregation operators	51
Information measures	32
Hybrid MCDM	16
TOPSIS	13
WASPAS	3
COPRAS	1
SWARA	1
VIKOR	3
ELECTRE	4
TODIM	3
QUALIFLEX	3
GRA	12
MULTIMOORA	1
PROMETHEE	1
Projection method	5
Other	14
Total	163

Source(s): Table created by author

Table 1.
Summary of
applications of the
MCDM techniques
([Peng and Dai, 2020](#))

Although, owing to the high dependency of this approach on the expert opinion and the tendency of experts to verbally express an opinion, the use of neutrosophic sets can remedy deficiencies such as indeterminacy, vagueness and uncertainty of verbal expressions or lack of information. But there is no development in the structure of the ISM approach and the next steps continue with binary numbers. In another study, [Dohale et al. \(2023\)](#) analyzed the enablers of the circular supply chain using NISM. They have used neutrosophic sets to obtain a neutrosophic judgment matrix. Although the novelty of this study is highlighting the NISM, in the early stages, they denitrified the judgment matrix in order to obtain crisp values for subsequent analyses. Although these two studies entered the neutrosophic space into ISM and MICMAC, But in these studies only experts' judgments are considered as neutrosophic sets and these conditions are not involved in the process of algorithms.

3. Introduction to neutrosophic sets and single-valued triangular neutrosophic numbers

In this section, the basic definitions from the neutrosophic sets and triangular neutrosophic numbers are presented, required for the following:

Definition 1. [Smarandache \(2004\)](#) proposed the concept of neutrosophic set N in the universal set X in which truth membership function $T_N(x)$, indeterminacy membership function $F_N(x)$, falsity membership function $I_N(x)$ are independently determined. These functions are defined as real standard or real non-standard subsets of X in the form of $T_N(x) =]^-0, 1^+]$, $F_N(x) =]^-0, 1^+]$ and $I_N(x) =]^-0, 1^+]$. Since there is no constraint on the sum of these functions, it is:

$$0^- \leq \sup T_N(x) + \sup I_N(x) + \sup F_N(x) \leq 3^+$$

Definition 2. [Smarandache \(2004\)](#) Let X be a universal set, then the single-valued neutrosophic set U is defined as:

$$U = \{ \langle x, T_U(x), I_U(x), F_U(x) \rangle : x \in X \}, \text{ Where } T_U(x) : X \rightarrow [0, 1], F_U(x) : X \rightarrow [0, 1], I_U(x) : X \rightarrow [0, 1], \text{ and } 0 \leq T_U(x) + I_U(x) + F_U(x) \leq 3 \text{ for } x \in X.$$

Definition 3. If $\alpha_a, \theta_a, \beta_a \in [0, 1]$ and $a_1, a_2, a_3 \in R$ such that $a_1 \leq a_2 \leq a_3$, then a single-valued triangular neutrosophic number (SVTrNN) is defined as $\tilde{a} = \langle (a_1, a_2, a_3); \alpha_a, \theta_a, \beta_a \rangle$. Its truth membership function, indeterminacy membership function and falsity membership function are defined as follows ([Deli and Subas, 2014](#)):

$$T_a^-(x) = \begin{cases} \alpha_a = \left(\frac{x - a_1}{a_2 - a_1} \right), & \text{if } a_1 \leq x \leq a_2 \\ \alpha_a, & \text{if } x = a_2 \\ \alpha_a = \left(\frac{a_3 - x}{a_3 - a_2} \right) & \text{if } a_2 \leq x \leq a_3 \\ 0 & \end{cases} \quad (1)$$

$$I_a^-(x) = \begin{cases} \left(\frac{a_2 - x + \theta_a^-(x - a_1)}{a_2 - a_1} \right) & \text{if } a_1 \leq x \leq a_2 \\ \theta_a^- & \\ \left(\frac{x - a_2 + \theta_a^-(a_3 - x)}{a_3 - a_2} \right) & \\ 1 & \end{cases} \quad (2)$$

$$F_a^-(x) = \begin{cases} \left(\frac{a_2 - x + \beta_a^-(x - a_1)}{a_2 - a_1} \right) & \\ \beta_a^- & \\ \left(\frac{x - a_2 + \beta_a^-(a_3 - x)}{a_3 - a_2} \right) & \\ 1 & \end{cases} \quad (3)$$

where α_a^- denotes maximum truth degree and θ_a^-, β_a^- denote minimum indeterminacy and minimum falsity degrees.

Definition 4. Şahin et al. (2018) Let $\tilde{a} = \langle (a_1, a_2, a_3); \alpha_a^-, \theta_a^-, \beta_a^- \rangle$ and $\tilde{b} = \langle (b_1, b_2, b_3); \alpha_b^-, \theta_b^-, \beta_b^- \rangle$ be two SVTrNN, then:

$$1. \tilde{a} + \tilde{b} = \langle (a_1 + b_1, a_2 + b_2, a_3 + b_3); \alpha_a^- \wedge \alpha_b^-, \theta_a^- \vee \theta_b^-, \beta_a^- \vee \beta_b^- \rangle \quad (4)$$

$$2. \tilde{a} - \tilde{b} = \langle (a_1 - b_3, a_2 - b_2, a_3 - b_1); \alpha_a^- \wedge \alpha_b^-, \theta_a^- \vee \theta_b^-, \beta_a^- \vee \beta_b^- \rangle \quad (5)$$

3. The Hamming distance between two SVTrNN can be defined as follow:

$$d(\tilde{a}, \tilde{b}) = \frac{1}{12} \left(|a_1(2 + \alpha_a^- - \theta_a^- - \beta_a^-) - b_1(2 + \alpha_b^- - \theta_b^- - \beta_b^-)| + |a_2(2 + \alpha_a^- - \theta_a^- - \beta_a^-) - b_2(2 + \alpha_b^- - \theta_b^- - \beta_b^-)| + |a_3(2 + \alpha_a^- - \theta_a^- - \beta_a^-) - b_3(2 + \alpha_b^- - \theta_b^- - \beta_b^-)| \right) \quad (6)$$

and

4. Deneutrosophication of SVTrNN can be defined as follow:

$$D\langle \tilde{a} \rangle = \frac{(a_1 + a_2 + a_3)}{3} + (\alpha_a^- - \theta_a^- - \beta_a^-) \quad (7)$$

5. Intersection between two SVTrNN can be defined as follow

$$\tilde{a} \cap \tilde{b} = \left(\max(a_1, b_1), \frac{a_2 + b_2}{2}, \min(a_3, b_3); \min(\alpha_a^-, \alpha_b^-), \max(\theta_a^-, \theta_b^-), \max(\beta_a^-, \beta_b^-) \right) \quad (8)$$

4. ISM

Interpretive structural modeling is a means of identifying interrelationships between various elements of a system and establishing a well-structured hierarchy of these elements. This modeling includes the following steps (Kaswan and Rathi, 2019; Warfield, 1974b):

- (1) To identify factors relevant to the concept (C): Through a literature review and information exchange with the experts of the field, these factors are identified.
- (2) To form structural self-interactive matrix (SSIM): Based on pairwise relationships, four symbols are used to describe the type of relationships between two factors (i, j). V, is used if factor " i " influences or reaches to factor " j ". A, is used if factor " j " reaches to factor " i ". X, is used if " i " and " j " reaches to each other. O, is used if both factors are unrelated.
- (3) To construct reachability matrix: By converting SSIM matrix into binary matrix, initial reachability matrix is obtained according to the following equations.

For (i, j), if it is V in SSIM then in reachability matrix (i, j) becomes "1" and (j, i) becomes "0".

For (i, j), if it is A in SSIM then in reachability matrix (i, j) becomes "0" and (j, i) becomes "1".

For (i, j), if it is X in SSIM then in reachability matrix (i, j) becomes "1" and (j, i) becomes "1".

For (i, j), if it is O in SSIM then in reachability matrix (i, j) becomes "0" and (j, i) becomes "0".

By modifying deficiencies and gaps in the initial reachability matrix, the final reachability matrix is obtained.

- (4) To level factors: Using the final reachability matrix, reachability and antecedent sets are derived for each factor. The reachability set includes the factor itself and the influential factors in achieving it. The antecedent set includes the factor itself and the factors influenced by its achievement. The intersection of these two sets is determined for all factors and if the intersection set is equal to the reachability set, the corresponding factor is considered to be in the first level and at the top of ISM hierarchy. These steps are repeated for the second factor and the iterations continue to determine the level of all factors.
- (5) To derive the digraph: According to the determined levels for the factors, a digraph of the relationships is derived.
- (6) To evaluate the model: Regarding any conceptual deficiency in the model, the necessary modifications are made.

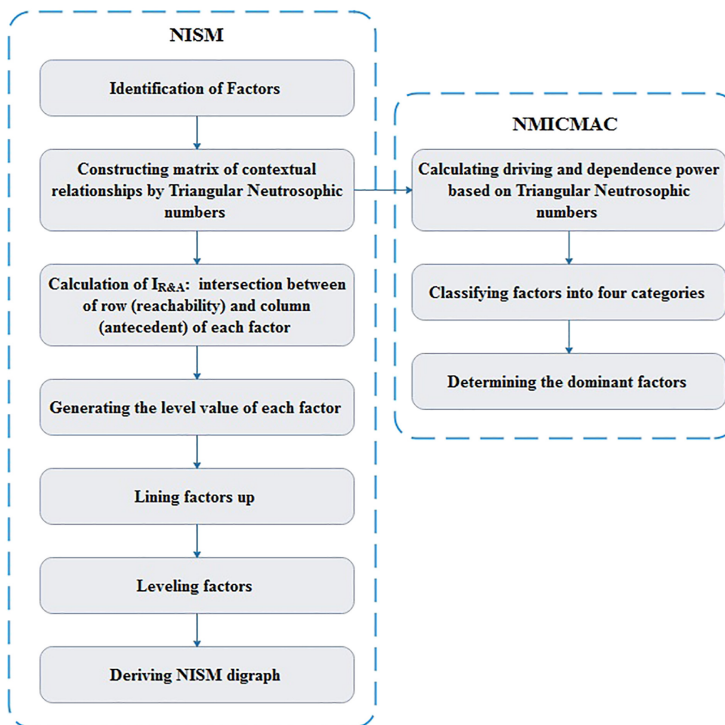
5. MICMAC analysis

MICMAC is a way to classify factors based on their driving and dependence power. The output of the final reachability matrix of ISM method is inputted to the MICMAC analysis. The ISM approach is used to create a factor ranking model, while the MICMAC analysis method is used to define the influence of the driving and the dependence between the factors (Khan *et al.*, 2022). The MICMAC algorithm follows these steps (Kaswan and Rathi, 2019; Wang *et al.*, 2018).

- (1) To calculate the driving and dependence effects. The driving power of each factor is obtained from the sum of its corresponding rows in the final reachability matrix and the dependence power of each factor is obtained from the sum of its corresponding column in the final reachability matrix.
- (2) To classify the factors. Factors are divided into four categories based on their driving and dependence power (Aloini *et al.*, 2012; Wang *et al.*, 2018).
 - Autonomous factors: these factors have weak driving as well as dependence power
 - Dependent factors: these factors have weak driving and strong dependence power
 - Linkage factors: these factors have strong driving as well as dependence power
 - Driver factors: these factors have strong driving and weak dependence power
- (3) To determine the dominant factors.

6. Neutrosophic ISM (NISM)

The decisions of experts and authorities depend on their experience and knowledge, which can be formed under uncertain, vague and indeterminate circumstances. Regarding the determining role of expert opinions in ISM and MICMAC, these approaches can be combined with neutrosophic logic in order to make the decisions believed to be more real. The steps for ISM method and MICMAC analysis are presented in Figure 1.



Source(s): Figure created by author

Figure 1.
Framework of
neutrosophic ISM and
MICMAC

6.1 Identification of factors (Ci)

The ISM method evaluates the relationships between variables relevant to an issue and depicts hierarchical relationships between them according to the degree of their driving and dependence effect. **Therefore, the first step in this method is determining a problem with its goals and identifying the factors affecting the problem. Reviews of literature, in-depth interviews with experts, or questionnaire surveys are among the common methods of gathering information on the issue and determining its factors (C1, C2, ..., Cn).**

6.2 Constructing matrix of contextual relationships by SVTrNN

In this stage, a panel of several experts is invited to fill the questionnaire in order to construct the contextual relationship matrix through pairwise comparison of the variables by the use of linguistic terms (very high, high, medium, low, etc.). **Then, SVTrNN substituted for linguistic variables obtained from the experts' opinions in the contextual relationship matrix. Triangular numbers are well suited for converting linguistic variables to small numbers. If several opinions are used to determine interrelations between variables, the final relationship matrix (YF) can be achieved by selecting the proposed relationship mode. By substitution of the neutrosophic numbers, the neutrosophic relationship matrix (YN) is obtained.**

$$Y_F = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{bmatrix} \quad (9)$$

$$\eta_{ij} = \langle (a_{ij}, b_{ij}, c_{ij}); \alpha_{ij}, \theta_{ij}, \beta_{ij} \rangle \quad (10)$$

$$Y_N = \begin{bmatrix} \eta_{11} & \eta_{12} & \cdots & \eta_{1n} \\ \eta_{21} & \eta_{22} & \cdots & \eta_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \eta_{n1} & \eta_{n2} & \cdots & \eta_{nn} \end{bmatrix} \quad (11)$$

6.3 Calculation of IR&A

In this stage, Reachability Set (S_R) and Antecedent Set (S_A) are formed for each factor using [Equations 12 and 13](#). With regard to the concept and logic of ISM operations, neutrosophic numbers, and [formula 8](#), the intersection (IR&A) of (S_R) and (S_A) is calculated in [equation 14](#), as a set of neutrosophic numbers ω_{ij} for each factor.

$$S_R(i) = \{ \eta_{i1} \quad \eta_{i2} \quad \cdots \quad \eta_{in} \} \quad (12)$$

$$S_A(i) = \{ \eta_{1i} \quad \eta_{2i} \quad \cdots \quad \eta_{ni} \} \quad (13)$$

$$I_{R\&A}(i) = \{ \eta_{i1} \cap \eta_{1i} \quad \eta_{i2} \cap \eta_{2i} \quad \cdots \quad \eta_{in} \cap \eta_{ni} / i \neq j \} \quad (14)$$

$$\omega_{ij} = \eta_{ij} \cap \eta_{ji}$$

$$= \left\langle \left(\max(a_{ij}, a_{ji}), \frac{b_{ij} + b_{ji}}{2}, \min(c_{ij}, c_{ji}) \right); \min(\alpha_{ij}, \alpha_{ji}), \max(\theta_{ij}, \theta_{ji}), \max(\beta_{ij}, \beta_{ji}) \right\rangle$$

$$\omega_{ij} = \langle (e_{ij}, f_{ij}, h_{ij}); \rho_{ij}, \delta_{ij}, \epsilon_{ij} \rangle$$

6.4 Generating the level value of each factor

In this stage, the level value (LV) of each factor is obtained from the distance between IR&A and Reachability (SR) sets using Hamming distance formula (Equation 7) for symmetric triangular fuzzy numbers (Equation 17).

$$d_{ij} = d(\overline{\eta_{ij}}, \overline{\omega_{ij}}) = \frac{1}{12} \left(\left| a_{ij}(2 + \alpha_{ij} - \theta_{ij} - \beta_{ij}) - e_{ij}(2 + \rho_{ij} - \delta_{ij} - \varepsilon_{ij}) \right| + \left| b_{ij}(2 + \alpha_{ij} - \theta_{ij} - \beta_{ij}) - f_{ij}(2 + \rho_{ij} - \delta_{ij} - \varepsilon_{ij}) \right| + \left| c_{ij}(2 + \alpha_{ij} - \theta_{ij} - \beta_{ij}) - h_{ij}(2 + \rho_{ij} - \delta_{ij} - \varepsilon_{ij}) \right| \right) \quad (16)$$

$$LV(C_i) = \sum_i d_{ij} \quad (17)$$

6.5 Lining factors up

By lining the factors up on the basis of LV in descending order, F set is composed of a series of factors corresponding to their importance (Relationship 18).

$$F = \{f_1, f_2, \dots, f_n / LV(f_i) \leq LV(f_{i+1})\} \quad (18)$$

6.6 Leveling factors

Factors are leveled on the base of LV distance between them. This distance is determined with the help of experts' opinions under the problem conditions. Then, factor leveling is done following the algorithm presented in Figure 2.

6.7 Deriving NISM digraph

In this stage, the authors derive a digraph considering the level of each factor. Firstly, the first level factors are positioned at the top of the hierarchy. Then, the highest LV of the second level factors is selected and compared with the mean LV of the first level factors. If it is higher than the mean, it is entered to the next level with no interference with the prior level; otherwise, it is entered to the next level interfering with the prior level. Figure 3 shows the algorithm for deriving the digraph.

7. Neutrosophic MICMAC (NMICMAC)

7.1 Calculating driving and dependence power of factors and forming MICMAC matrix

In this step, the driving power and dependence power of each factor are calculated using formula 1. Driving Power (PDR) and Dependence Power (PDE) are respectively obtained from the sum of numbers in the row and the sum of numbers in the column corresponding to each factor. Then, the neutrosophic numbers are converted to definite numbers for which MICMAC matrix is drawn.

$$P_{DR}(i) = \sum_j \eta_{ij} = \left\langle \left(\sum_j a_{ij}, \sum_j b_{ij}, \sum_j c_{ij} \right); \cap_j \alpha_{ij}, \cup_j \theta_{ij}, \cup_j \beta_{ij} \right\rangle \quad (19)$$

$$P_{DE}(i) = \sum_i \eta_{ij} = \left\langle \left(\sum_i a_{ij}, \sum_i b_{ij}, \sum_i c_{ij} \right); \cap_i \alpha_{ij}, \cup_i \theta_{ij}, \cup_i \beta_{ij} \right\rangle \quad (20)$$

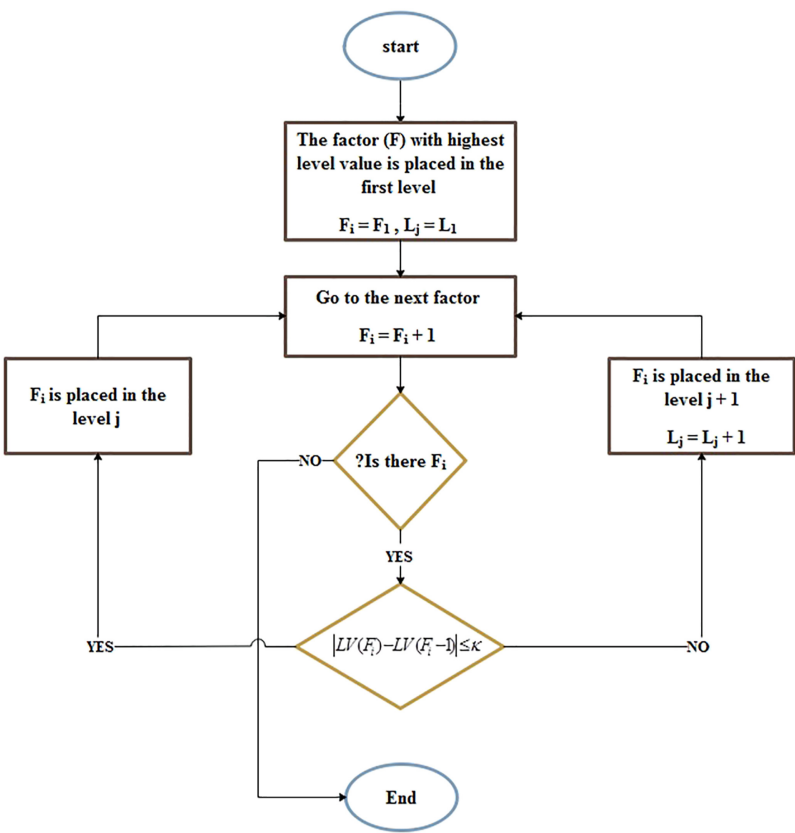


Figure 2.
Factor leveling
algorithm in NISM

Source(s): Figure created by author

8. Numerical example

8.1 Identification of the factors

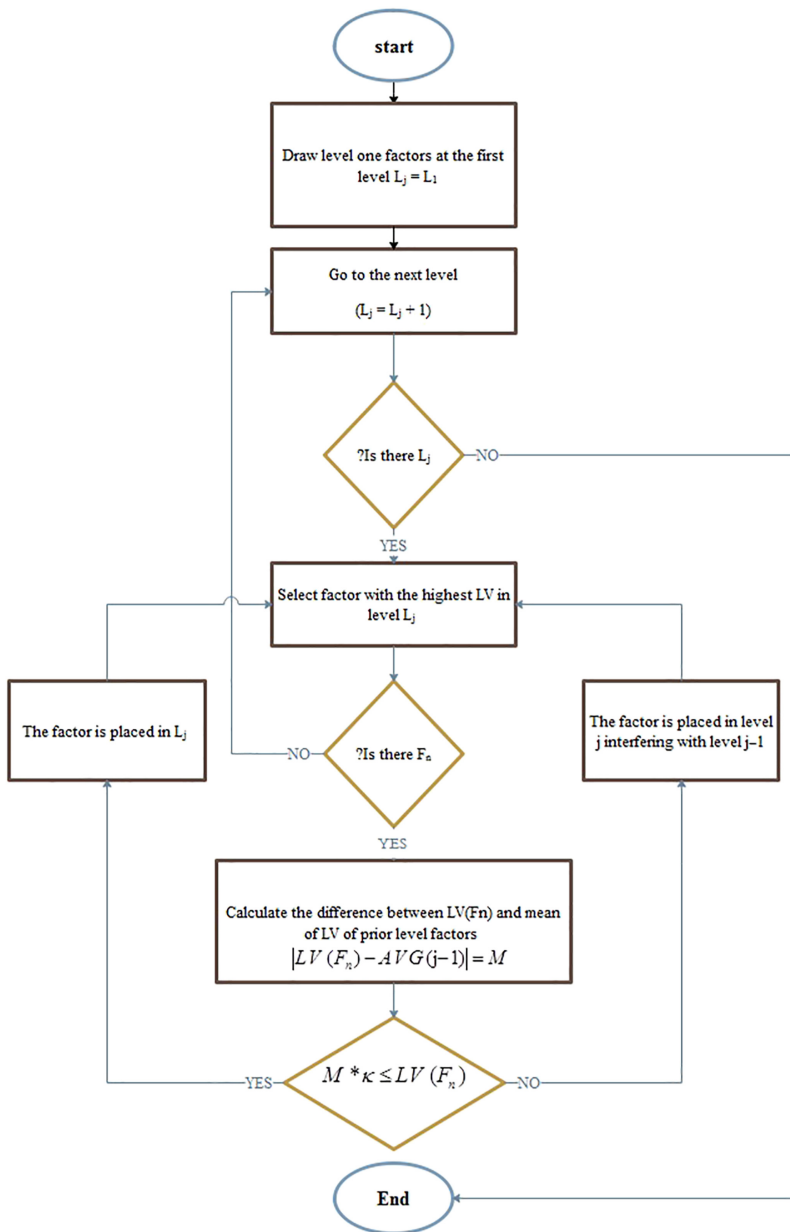
The goal is to level the factors affecting the selection of suppliers and identify dependent and driver factors in the system with this assumption that there are 13 factors affecting the selection of suppliers. These factors are: C1: quality, C2: price, C3: delivery, C4: service, C5: information sharing, C6: attitude, C7: managed inventory, C8: agility, C9: information technology, C10: transport, C11: financial situation, C12: top management commitment, C13: strategic goals.

8.2 Formation of contextual relationship matrix

The final matrix of pairwise relationships according to the linguistic variables and consensus of opinions is shown in Table 2 using the SVTrNNs associated with the linguistic expressions (Table 3), the table of neutrosophic pair relationships is drawn (Table 4).

8.3 Calculation of IR&A

In this stage, Reachability and Antecedent sets are developed for 13 factors, and their intersection is derived from Equation 15. If there is no intersection, the SVTrNN (0,0,0;1,0,0) are considered to continue the iterations. An example of 13 factors shows in Table 4. IR&A calculation for factor C1 shows in Table 5.



Source(s): Figure created by author

Figure 3.
NISM digraph
algorithm

8.4 Calculation of level value (LV)

Using Equation 16 and 17, the level value of 13 factors affecting the supplier selection is obtained and the results are shown in Table 6.

Table 2.
Final matrix of
relationships
according to the
linguistic variables

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
C1	–	H	NO	NO	NO	M	NO	H	NO	NO	NO	NO	NO
C2	H	–	H	H	NO	NO	H	NO	M	NO	NO	NO	NO
C3	NO	H	–	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
C4	M	H	NO	–	NO	NO	NO	NO	NO	NO	NO	NO	NO
C5	H	M	H	H	–	NO	H	NO	NO	NO	NO	NO	NO
C6	NO	NO	VH	H	M	–	H	VH	H	M	NO	NO	NO
C7	NO	NO	VH	NO	NO	NO	–	VH	NO	NO	NO	NO	NO
C8	VH	VH	H	M	H	NO	VH	–	NO	NO	NO	NO	NO
C9	L	L	VH	VH	VH	H	H	VH	–	VH	NO	NO	NO
C10	NO	H	H	L	NO	NO	NO	VH	NO	–	NO	NO	NO
C11	H	H	H	H	NO	NO	H	H	H	H	–	H	H
C12	VH	L	VH	H	H	VH	M	H	VH	L	VH	–	VH
C13	H	H	H	H	H	VH	H	M	VH	H	NO	NO	–
Source(s): Table created by author													

Table 3.
Triangular
neutrosophic scale for
comprise matrix

Linguistic term	Triangular neutrosophic scale
No influence (NO)	(0.1,0.2,0.3;0.5,0.1,0.3)
Low influence (L)	(0.2,0.3,0.4;0.8,0.2,0.3)
Medium influence (M)	(0.4,0.5,0.6;0.7,0.3,0.2)
High influence (H)	(0.6,0.7,0.8;0.8,0.3,0.5)
Very high influence (VH)	(0.7,0.8,0.9;0.9,0.3,0.5)
Source(s): Table created by author	

8.5 Lining factors up

13 factors influencing the selection of suppliers are arranged in descending order based on the obtained LVs.

C1, C3, C2, C4, C7, C10, C6, C8, C5, C9, C11, C12, C13.

8.6 Leveling factors

Using the presented algorithm in Figure 2 and the arranged factors, leveling is done. In this example, according to the experts' opinions and neutrosophic numbers defined for linguistic variables (Equation 10), the authors consider the value $c_{ij} - a_{ij}$ as κ . On this basis, the initial leveling of the factors is done and the results are represented in Table 7.

8.7 Deriving the digraph

Considering $\kappa = 0.2$ and the presented algorithm in Figure 3, the digraph of factors influencing the selection of suppliers is drawn (Figure 4). The criteria 6, 9 and 11 interfere with the lower level and their importance will be more than the same level factors.

8.8 N-MICMAC

The sum of row neutrosophic numbers for each factor shows its neutrosophic driving power (Equation 19) and the sum of column neutrosophic numbers shows its neutrosophic dependence power (Equation 20). Results are summarized in Table 8 and MICMAC matrix is shown in Figure 5.

C1	C2	C3	C4	C5	C6
C1 (1,1,1,0,0)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,4,0,5,0,6,0,7,0,3,0,2)
C2 (0,6,0,7,0,8,0,8,0,3,0,5)	(1,1,1,1,0,0)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C3 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,6,0,7,0,8,0,8,0,3,0,5)	(1,1,1,1,0,0)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C4 (0,4,0,5,0,6,0,7,0,3,0,2)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(1,1,1,1,0,0)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C5 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(1,1,1,1,0,0)	(0,1,0,2,0,3,0,5,0,1,0,3)
C6 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,1,0,2,0,3,0,5,0,1,0,3)
C7 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C8 (0,7,0,8,0,9,0,9,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)
C9 (0,2,0,3,0,4,0,8,0,2,0,3)	(0,2,0,3,0,4,0,8,0,2,0,3)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)
C10 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,2,0,3,0,4,0,8,0,2,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C11 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C12 (0,7,0,8,0,9,0,9,0,3,0,5)	(0,2,0,3,0,4,0,8,0,2,0,3)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)
C13 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)

C7	C8	C9	C10	C11	C12	C13
C1 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C2 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C3 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C4 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C5 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C6 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C7 (1,1,1,0,0)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C8 (0,7,0,8,0,9,0,9,0,3,0,5)	(1,1,1,1,0,0)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C9 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C10 (0,1,0,2,0,3,0,5,0,1,0,3)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,0,2,0,3,0,5,0,1,0,3)
C11 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,1,1,0,0)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)
C12 (0,4,0,5,0,6,0,7,0,3,0,2)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,2,0,3,0,4,0,8,0,2,0,3)	(1,1,1,1,0,0)	(0,7,0,8,0,9,0,9,0,3,0,5)
C13 (0,6,0,7,0,8,0,8,0,3,0,5)	(0,4,0,5,0,6,0,7,0,3,0,2)	(0,7,0,8,0,9,0,9,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,6,0,7,0,8,0,8,0,3,0,5)	(0,1,0,2,0,3,0,5,0,1,0,3)	(0,1,1,1,0,0)

Source(s): Table created by author

Table 4.
Contextual
relationship matrix
by SVTRNNs

C1	C2	C3	C4	C5	C6	C7
Reachability	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.4,0.5,0.6,0.7,0.3,0.2)	(0.1,0.2,0.3,0.5,0.1,0.3)
Antecedent	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.4,0.5,0.6,0.7,0.3,0.2)	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)
IR&A	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.0,0.1,0.0)	(0.0,0.1,0.0)	(0.0,0.1,0.0)	(0.1,0.2,0.3,0.5,0.1,0.3)
C1	C8	C9	C10	C11	C12	C13
Reachability	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)
Antecedent	(0.7,0.8,0.9,0.9,0.3,0.5)	(0.2,0.3,0.4,0.8,0.2,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.6,0.7,0.8,0.8,0.3,0.5)	(0.7,0.8,0.9,0.9,0.3,0.5)	(0.6,0.7,0.8,0.8,0.3,0.5)
IR&A	(0.7,0.75,0.8,0.8,0.3,0.5)	(0.2,0.25,0.3,0.5,0.2,0.3)	(0.1,0.2,0.3,0.5,0.1,0.3)	(0.0,0.1,0.0)	(0.0,0.1,0.0)	(0.0,0.1,0.0)

Source(s): Table created by author

Criteria	Level value	An extended neutrosophic ISM and MICMAC 773 Table 6. Level value of supplier selection factors
C1	0.85	
C2	0.96	
C3	0.95	
C4	1.03	
C5	2.2	
C6	1.75	
C7	1.16	
C8	2.07	
C9	2.86	
C10	1.61	
C11	3.18	
C12	3.57	
C13	3.78	
Source(s): Table created by author		

Criteria	Level
C12, C13	1
C11	2
C9	3
C5, C8	4
C6, C10	5
C1, C2, C3, C4, C7	6
Source(s): Table created by author	

Table 7.
Initial leveling of
supplier selection
factors

9. Discussion

ISM is a technique that helps decision-makers better understand the complex relationships between the various factors influencing a process. It presents a hierarchical model based on the importance of factors. MICMAC is also the driver-dependence diagram that shows the degree of independence and dependence of factors (Singh *et al.*, 2019). Although these two methods are widely used and effective techniques in soft studies, they have limitations that are tried to be dealt with in this study.

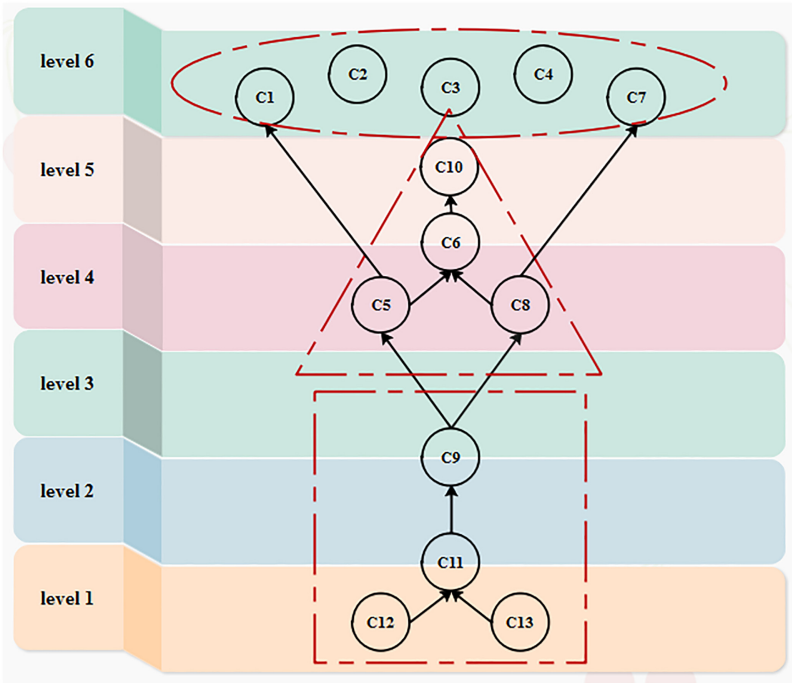
9.1 Theoretical contributions

In practice, managers and decision-makers can use group knowledge and experience and apply these techniques to depict complex situations in a simple way and make the necessary decisions. But they face with two challenges in the process of using these techniques:

- (1) Vague and inconsistent information of real situation
- (2) Limited knowledge and lack of accurate evaluation of pairwise comparisons judgment

Despite these challenges, Classic ISM and MICMAC cannot be accurate. Therefore, in the background of research, attempts have been made to provide solutions to address these challenges (Bianco *et al.*, 2021). In this regard, although fuzzy and intuitionistic ISM and MICMAC are introduced, in fuzzy only the truth degree is hold and in intuitionistic, the truth and falsity degrees are considered. Therefore, these two approaches are not able to deal with

Figure 4.
NISM digraph of
supplier selection
factors



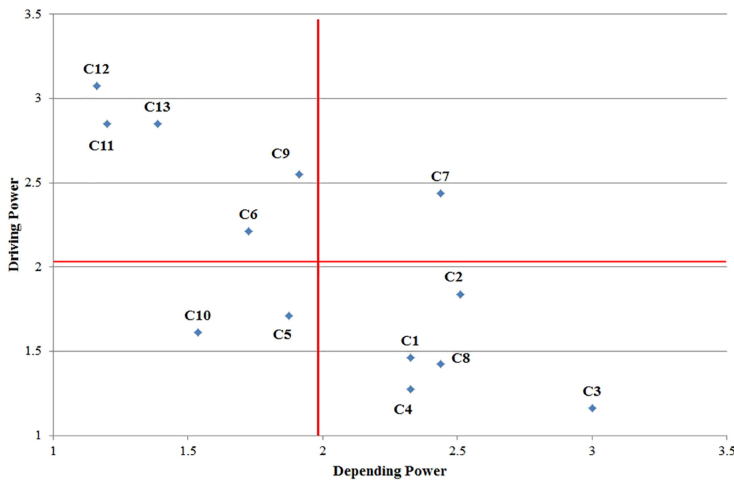
Source(s): Figure created by author

Table 8.
Neutrosophic
dependence and
driving power of
supplier selection
factors

Criteria	N-P _{DR}	N-P _{DE}	Crisp-P _{DR}	Crisp-P _{DE}
C1	(3.5,4.7,5.9;0.5,0.3,0.5)	(5.8,7.8,2;0.5,0.3,0.5)	1.4625	2.325
C2	(4.5,5.7,6.9;0.5,0.3,0.5)	(6.3,7.5,8.7;0.5,0.3,0.5)	1.8375	2.5125
C3	(2.7,3.9,5.1;0.5,0.3,0.5)	(7.6,8.8,10;0.5,0.3,0.5)	1.1625	3
C4	(3.4,2.5,4;0.5,0.3,0.5)	(5.8,7.8,2;0.5,0.3,0.5)	1.275	2.325
C5	(4.5,5.2,6.4;0.5,0.3,0.5)	(4.6,5.8,7;0.5,0.3,0.5)	1.7125	1.875
C6	(5.5,6.7,7.9;0.3,0.3,0.5)	(4.2,5.4,6.6;0.5,0.3,0.5)	2.2125	1.725
C7	(3.4,4.6,5.8;0.5,0.3,0.5)	(6.1,7.3,8.5;0.5,0.3,0.5)	2.4375	2.4375
C8	(5.3,6.5,7.7;0.5,0.3,0.5)	(6.1,7.3,8.5;0.5,0.3,0.5)	1.425	2.4375
C9	(6.4,7.6,8.8;0.5,0.3,0.5)	(4.7,5.9,7.1;0.5,0.3,0.5)	2.55	1.9125
C10	(3.9,5.1,6.3;0.5,0.3,0.5)	(3.7,4.9,6.1;0.5,0.3,0.5)	1.6125	1.5375
C11	(7.2,8.4,9.6;0.5,0.3,0.5)	(2.8,4.5,2;0.5,0.3,0.5)	2.85	1.2
C12	(7.8,9.10,20.5;0.3,0.5)	(2.7,3.9,5.1;0.5,0.3,0.5)	3.075	1.1625
C13	(7.2,8.4,9.6;0.5,0.3,0.5)	(3.3,4.5,5.7;0.5,0.3,0.5)	2.85	1.3875

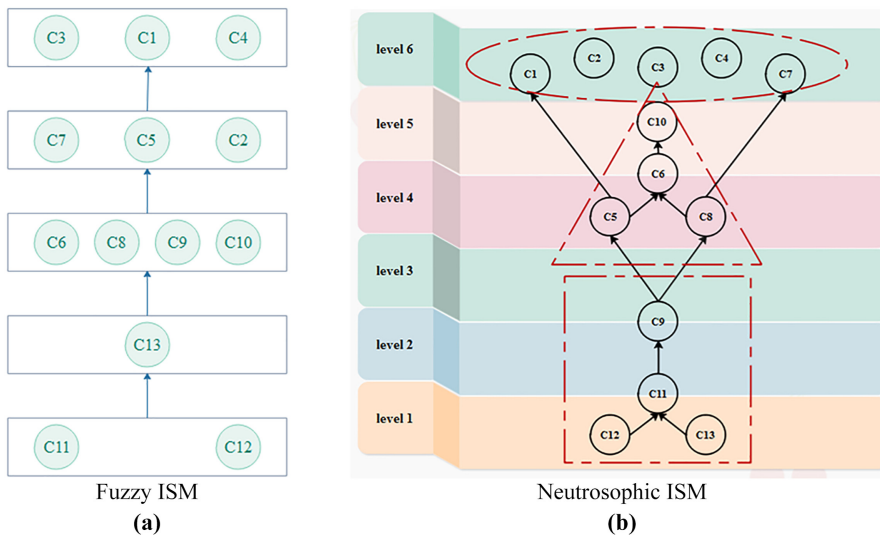
Source(s): Table created by author

vague and inconsistent and complete conversion of linguistic variables. While in most of these studies verbal expressions have been used only to obtain the reachability matrix, then by converting the degrees of membership into crisp numbers, the main stages of the ISM method have continued with binary numbers (Bhosale and Kant, 2016; Khan *et al.*, 2021; Khatwani and Prakash, 2015; Shanker and Barve, 2021). Fuzzy ISM model of the proposed example is shown in Figure 6a.



Source(s): Figure created by author

Figure 5.
NMICMAC of supplier
selection factors



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Figure 6.
Fuzzy and
neutrosophic ISM
models

Recently, in a study, [Karadayi-Usta \(2022\)](#) proposed a NISM to examine the relationship of the medical tourism services' barriers. In another study, [Dohale et al. \(2023\)](#) analyzed the enablers of the circular supply chain using NISM. These studies have attempted to determine hidden patterns of relationships between factors using neutrosophic sets. But, like many researches related to fuzzy ISM, neutrosophic sets were used only in the initial stages and to achieve the reachability matrix. Although, owing to the high dependency of this approach on the expert opinion and the tendency of experts to verbally express an opinion, the use of neutrosophic sets can remedy deficiencies such as indeterminacy, vagueness and uncertainty of verbal

expressions or lack of information. But there is no development in the structure of the ISM approach and the next steps continue with binary numbers.

In this study, the neutrosophic sets have been used to overcome the mentioned challenges and eliminate the previous shortcomings, where each pairwise comparison judgment is represented as SVTrNN. SVTrNN can provide judgmental information more effectively. Neutrosophic sets cover not only truth membership function but also indeterminacy membership function and falsity membership function.

These features help to better show, redundancy, inconsistency and contradiction in addition to ambiguity and uncertainty in real-world information. NISM and NMICMAC not only improve in theory and have the ability to more accurately reflect the real situation but also help managers to make decisions by minimizing bias resulting from individual judgments in practice. In this study, the development of the model is continued until the final stages of calculations with neutrosophic numbers, and only in the final stage the results are crisply presented. In addition to solving the above challenges in the special algorithm presented in this paper, not only the factors of process are leveled, but at each level the factors are lined up and their importance is determined. However, traditional ISM only deals with the leveling of factors and factors at every level have equal priority. For example, although C6 and C10 are on the same level in both the fuzzy and neutrosophic models (Figures 6a and 6b), in the neutrosophic model C6 is closer to the previous level and has a higher degree of importance due to the obtained LV. In this way, by lining up the factors using LV in the following order (C1, C3, C2, C4, C7, C10, C6, C8, C5, C9, C11, C12, C13), the priority of the factors at each level can be easily determined. According to the logic of LV, these numbers can be considered as the weight of factors and can be used for example in other calculations, including in other MCDM techniques. Therefore, decision makers and managers may be hesitant to make decisions about factors at one level. But the lining up and level value of factors can make the decision easier and more accurate for them. Finally, this new algorithm can affect the final decisions and selection of factors.

9.2 Managerial and practical contributions

The research presented in this article has important implications for both practical applications and managerial decision-making. The study addresses the need for a systematic framework for utilizing ISM and MICMAC methods in complex and uncertain conditions, where ambiguity is likely to arise. By leveraging the unique features of the neutrosophic space, the proposed approach allows for a more flexible and accurate evaluation of the problem domain. This can lead to more realistic and informed results, as the proximity of real-world conditions to the assumptions of the neutrosophic theory enables a better alignment with the problem context. In addition, the proposed framework facilitates the complex judgment stage of decision-making, which is often prone to introducing ambiguity and uncertainty in the process.

Given the critical role of decision-making in management, the findings of this research can significantly aid decision-makers in making more informed and accurate decisions, based on a more realistic understanding of the problem context. The results also provide a foundation for further research, enabling the conceptualization of other MCDM methods in the neutrosophic space, leading to further advancements in the field. Overall, the proposed approach and framework can be highly beneficial in dealing with complex and uncertain problems, enabling decision-makers to navigate ambiguity and make informed decisions.

10. Conclusion

Interpretive structural modeling (ISM) was first introduced by Warfield (1974a) as a method to determine the complex interrelations between factors affecting a system using experts' opinions. Subsequently fuzzy theory was introduced by Zadeh (1965) and used in some other studies, including

ambiguity and uncertainty. However, fuzzy theory was not effective in expressing contradiction and inconsistency between opinions. To address this limitation, Smarandache (1999) proposed neutrosophic logic as a method for dealing with indeterminate, vague and inconsistent information.

In this study, we propose a model for leveling factors to deal with different conditions. The matrix of interrelations between factors influencing the system was obtained using single-valued triangular neutrosophic number (SVTrNN) related to the linguistic variables for which ISM and MICMAC were processed. We provide the general algorithm of Neutrosophic Interpretive Structural Modeling (NISM) and neutrosophic MICMAC (NMICMAC), as well as the algorithms of factor leveling and digraph.

Our proposed model aims to process neutrosophic data while satisfying the neutrosophic condition and preserving the effect of neutrosophic numbers on the model until the last step. Factors are not only leveled but also lined up, with each factor's distance from and interference with the prior level factors considered in the digraph, showing its relative importance to the same level factors. Furthermore, corresponding value to the opinion of experts, under special circumstances, it is possible to determine leveling in accordance with the condition.

This study presents a novel model for leveling factors that can be used to handle contradictory, inconsistent, uncertain and vague conditions affecting a system. The obtained Level Value (LVs) in this method can also be used as weights for factors in multi-criteria decision-making methods. In future studies, this model can be applied to various fields to provide a more comprehensive and accurate analysis of complex interrelationships between factors affecting a system.

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